1st Place: Aspect 9 – Translate a problem into a standard mathematical representation or algorithm

- **Analysis**: This is the most central modeling step. Students must translate the profit calculation rule described in the text into a precise algebraic equation (standard mathematical representation).
- **Manifestation in the Question**: The key text is: "The normal selling price... includes a profit of 37.5%. The profit is calculated as a percentage of the wholesale price." This must be translated into:
  - **Profit** = Wholesale price w × 37.5% = $0.375w$
  - **Normal selling price** s = Wholesale price w + Profit = $w + 0.375w = 1.375w$
    Or its equivalent forms. Successfully completing this translation establishes the core criterion for judging the correctness of all formulas ($s = 1.375w$ or its equivalent transformations).

2nd Place: Aspect 4 – Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

- **Analysis**: This is the prerequisite for modeling. Students must carefully read and understand the textual definitions of "profit," "wholesale price," "normal selling price," and their percentage relationship in the problem.
- **Manifestation in the Question**: Students need to decode the two key statements: "includes a profit of 37.5%" and "profit is calculated as a percentage of the wholesale price." They need to understand that the base for the 37.5% is the wholesale price w, not the selling price s. This is the foundation for building the correct model; misunderstanding it would lead to all judgments being wrong.

3rd Place: Aspect 6 – Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

- **Analysis**: After understanding the context, the key mathematical objects (variables) and the quantitative relationships between them need to be abstracted.
- **Manifestation in the Question**: Students need to identify the two core variables: **wholesale price** w and **normal selling price** s. More importantly, they need to identify the **mathematical relationship** between them: the selling price is composed of the wholesale price plus a profit calculated based on the wholesale price.

4th Place: Aspect 5 – Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations

- **Analysis**: Students need to see that the profit calculation is based on a **linear growth** structural pattern.
- **Manifestation in the Question**: The relationship described is essentially: selling price = wholesale price × (1 + profit rate). This is a typical **multiplicative growth** pattern (percentage increase). Students need to recognize this pattern and use it to test the formulas. For example, they need to see that $s = w + 0.375$ represents an **additive constant** pattern, which does not fit the percentage growth pattern and is therefore incorrect.

5th Place: Aspect 1 – Select a mathematical description or a representation that describes a problem

- **Analysis**: After establishing the core relationship $s = 1.375w$, students need to use this relationship as a "ruler" to measure (choose/judge) whether each given formula is a correct representation of this relationship.
- **Manifestation in the Question**: For each formula, students need to judge whether it is algebraically equivalent to $s = 1.375w$. This requires them to select the correct mathematical description (the standard model) to evaluate alternative descriptions. For instance, judging whether $w = s - 0.375s$ is equivalent to $s = 1.375w$ (simplifying gives $w = 0.625s$, which does not match $w = s/1.375 \approx 0.727s$ derived from $s = 1.375w$, so it is wrong).

6th Place: Aspect 8 – Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

- **Analysis**: This problem can be immediately associated with known mathematical concepts and problem types such as "percentage increase," "relationship between cost, profit, and selling price," or "translating word problems into equations."
- **Manifestation in the Question**: If students are familiar with classic problems like "cost (or purchase price) increased by a certain percentage yields selling price," they can quickly recognize the structure of this problem and invoke the relevant solution approach (setting up the equation $s = (1 + p)w$).

7th Place: Aspect 12 – Identify the constraints, assumptions simplifications in a mathematical model

- **Analysis**: The model given in the problem is a simplification of real-world commercial pricing.
- **Manifestation in the Question**: Key assumptions in the model include: 1) Profit is calculated **solely** based on the wholesale price; 2) The profit rate is fixed at 37.5%; 3) The selling price is composed only of the wholesale price and profit, with no other costs or discounts. Students need to identify and accept these simplifying assumptions to judge the formulas within this framework. If a student considers that "profit might also be calculated based on selling price," that goes beyond the constraints of this problem's model.

8th Place: Aspect 10/11 – Use mathematical tools (variables, symbols) to describe mathematical structures and relationships

- **Analysis**: During the analysis and judgment process, students need to use algebraic symbols to express and manipulate relationships.
- **Manifestation in the Question**: To judge the formulas, students may need to use algebraic tools for derivation and verification. For example, deriving $w = s/1.375$ from $s = 1.375w$ and comparing it with the option $w = 0.625s$. Or manipulating an option to see if it yields $s = 1.375w$. This process is using algebraic symbols as a tool to clearly describe and test relationships.

第1位：方面 9 - 将问题转化为标准的数学表征或算法

- **分析**：这是本题最核心的建模步骤。学生必须将文字描述中的利润计算规则，转化为一个精确的代数方程（标准数学表征）。
- **题目体现**：关键的文字是："The normal selling price... includes a profit of 37.5%. The profit is calculated as a percentage of the wholesale price." 这需要被转化为：
  - **利润** = 批发价 w × 37.5% = $0.375w$
  - **正常售价** s = 批发价 w + 利润 = $w + 0.375w = 1.375w$
    或者其等价形式。成功完成这一翻译，就建立了判断所有公式对错的核心标准（$s = 1.375w$ 或其等价变形）。

第2位：方面 4 - 阅读、解码并理解陈述，以建立情境模型

- **分析**：这是建模的前提。学生必须仔细阅读并理解题目中关于"利润"、"批发价"、"正常售价"及其百分比关系的文字定义。
- **题目体现**：学生需要解码"includes a profit of 37.5%"和"profit is calculated as a percentage of the wholesale price"这两个关键陈述。他们需要理解37.5%的基数是批发价 w，而不是售价 s。这是建立正确模型的基础，误解会导致所有判断错误。

第3位：方面 6 - 识别并描述现实问题中的数学方面，包括识别重要变量

- **分析**：在理解情境后，需要从中抽象出关键的数学对象（变量）和它们之间的数量关系。
- **题目体现**：学生需要识别出两个核心变量：**批发价** w 和**正常售价** s。更重要的是，识别出它们之间的**数学关系**：售价由批发价加上基于批发价的利润构成。

第4位：方面 5 - 在问题或情境中识别数学结构（规律、关系、模式）

- **分析**：学生需要看出利润计算背后是一个**线性增长**的结构模式。
- **题目体现**：题目描述的关系本质上是：售价 = 批发价 × (1 + 利润率)。这是一个典型的**乘性增长**模式（百分比增长）。学生需要识别出这个模式，并用它来检验公式。例如，他们需要看出 $s = w + 0.375$ 是一个**加性常数**模式，不符合百分比增长的模式，因此是错误的。

第5位：方面 1 - 选择描述问题的数学描述或表征

- **分析**：在建立了核心关系 $s = 1.375w$ 之后，学生需要运用这个关系作为“标尺”，去衡量（选择判断）题目给出的每一个公式是否是该关系的正确表征。
- **题目体现**：对于每个公式，学生需要判断它是否与 $s = 1.375w$ 在代数上等价。这要求他们选择正确的数学描述（标准模型）来评估备选描述。例如，判断 $w = s - 0.375s$ 是否等价于 $s = 1.375w$（化简后是 $w = 0.625s$，这与从 $s = 1.375w$ 解出的 $w = s/1.375 \approx 0.727s$ 不符，因此错误）。

第6位：方面 8 - 识别与已知问题或数学概念的对应

- **分析**：这个问题可以立刻与“百分比增长”、“成本、利润、售价关系”或“将文字问题转化为方程”等已知的数学概念和问题类型对应起来。
- **题目体现**：学生如果熟悉“成本(或进价)增加一定百分比得到售价”这类经典问题，就能迅速识别出本题的结构，从而调用相关的解决思路（建立方程 $s = (1 + p)w$）。

第7位：方面 12 - 识别数学模型中的约束、假设与简化

- **分析**：题目给出的模型是对现实商业定价的简化。
- **题目体现**：模型中的关键假设包括：1) 利润**仅**基于批发价计算；2) 利润率是固定的37.5%；3) 售价仅由批发价和利润构成，没有其他成本或折扣。学生需要识别并接受这些简化假设，才能在此框架下判断公式。如果学生考虑“利润也可能基于售价计算”，那就超出了本题模型的约束。

第8位：方面 10/11 - 使用数学工具（变量、符号）描述数学结构和关系

- **分析**：在分析和判断过程中，学生需要使用代数符号来表述和操作关系。
- **题目体现**：为了判断公式，学生可能需要使用代数工具进行推导和验证。例如，从 $s = 1.375w$ 推导出 $w = s/1.375$，然后与选项 $w = 0.625s$ 比较。或者将选项变形，看是否能得到 $s = 1.375w$。这个过程就是在使用代数符号这一工具来清晰描述和检验关系。